

G H Raisoni College of Engineering and Management, Pune
(An Empowered Autonomous Institute affiliated to Savitribai Phule, Pune University, NAAC Accredited "A+" Grade)
T.Y B.Tech CSE (Data Science)
(TERM- I)
TAE 2- Winter 2026
23UDSPCL3508/23UDSPCP3508: Database Management Systems
(Max. Marks: 20)

All Third Year Data Science students are hereby informed that TAE 2 (20 Marks) for Database Management Systems (DBMS) will be evaluated through a Live Project Implementation, SQL Execution, and Technical Oral Presentation.

Continuity Clause (TAE 1 to TAE 2)

As outlined in TAE 1, this task is the direct continuation (Phase II) of your continuous evaluation:

- **Approved Domain & Design:** You **MUST** implement the exact relational schema, business rules, and domain approved during TAE 1.
- **No Topic Switching:** Changing database topics or database engines between TAE 1 and TAE 2 is **STRICTLY PROHIBITED**.

Execution & Technical Requirements

Students are required to carry out full database lifecycle implementation in their chosen target RDBMS engine (MySQL, PostgreSQL, Oracle, SQL Server, etc.):

1. **DDL & Schema Instantiation:**
 - Translate your 3NF/BCNF relational model from TAE 1 into executable SQL code (CREATE TABLE, ALTER TABLE).
 - Explicitly define Primary Keys, Foreign Keys (ON DELETE, ON UPDATE cascades), NOT NULL, UNIQUE, CHECK, and default constraints.
2. **Data Population & Synthetic/Raw Import:**
 - Populate your tables with a minimum of 100+ realistic records per core table (using mock generator scripts, Python Faker, or transformed raw datasets).
 - Ensure data consistency so that complex joins and query conditions yield non-trivial results.
3. **Advanced SQL Implementation:**
 - Write and execute queries demonstrating:
 - Complex multi-table Inner, Outer, and Self Joins.
 - Correlated Subqueries and Aggregate Functions with GROUP BY / HAVING.
 - At least 1 Stored Procedure with parameters for operational transactions.
 - At least 1 Trigger enforcing an automated business rule or dynamic auditing.
 - At least 2 Materialized/Virtual Views for business reporting.
4. **Performance Tuning & Indexing:**
 - Execute EXPLAIN or EXPLAIN ANALYZE on at least two slow/complex queries.
 - Create appropriate Single-Column or Composite Indexes (CREATE INDEX) to demonstrate query performance improvement before and after indexing.

Evaluation Components & Presentation Format

The evaluation for TAE 2 consists of two deliverables: a GitHub Repository/Documentation Submission and an In-Person Technical Presentation.

1. Presentation Structure (10 Minutes Total)

- Slide 1: Domain Overview & TAE 1 Recall (1 Min) — Brief introduction of the domain and final 3NF Relational Schema.
- Slide 2: System Architecture & Implementation Setup (1 Min) — Target RDBMS, tools used, and database topology.
- Live DBMS Demonstration (6 Mins):
 - Live terminal/GUI execution of SQL DDL, DML, Stored Procedures, and Triggers.
 - Demonstration of query plan optimisation using EXPLAIN with and without Indexes.
- Technical Q&A / Viva (2 Mins): Defence of indexing choices, ACID compliance, and concurrency controls.

2. Required GitHub / Drive Deliverables

Students must submit a clean directory containing:

- schema.sql (Table creation DDL with constraints).
- data.sql or populate.py (DML insert scripts or automated data generators).
- queries.sql (Complex queries, Views, Stored Procedures, Triggers, and EXPLAIN benchmarks).
- Presentation_Deck.pdf (Slide deck used during viva).

Marks Evaluation Breakdown for TAE 2 (Total: 20 Marks)

Q.No.	Question	Marks	CO Mapped	BL
1	Database Instantiation & Advanced SQL Implementation Construct database schema using DDL with all integrity constraints. Populate data (100+ rows per table). Implement complex multi-table joins, subqueries, views, stored procedures, and triggers reflecting TAE 1 business rules.	10	CO2	BL3
2	Performance Optimisation & Live Presentation / Defense Conduct query cost analysis using EXPLAIN. Implement indexes to optimize performance. Present project architecture and perform a live query execution & viva defense before the evaluation panel.	10	CO3	BL4

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